

ActInf Livestream #026.1 ~ “Bayesian Mechanics for Stationary Processes”

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hello everyone welcome to the active inference lab today it is august 10th 2021 and we're here in live stream number 26.1 discussing bayesian mechanics for stationary processes welcome to the active inference lab if it's your first time or not we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams so people can just raise their hand visually or in the jitsi if they want to speak this short link here has our calendar of live streams for the year and we're here in number 26 where today on the 10th and then next week on the 17th we're going to be continuing discussion on the bayesian mechanics for stationary processes paper and then we're going to launch into some other papers so take a look at that link if you want to learn more today in live stream number 26.1 the goal is going to be to learn and discuss thankfully and really appreciatively with some of the authors of the paper we're going to be discussing this bayesian mechanics for stationary processes paper and asking questions writing things down we want to look up over the coming week and just seeing what is uh fun to talk about in this paper so we'll start with just an introduction and a warm-up of the non-authors and then we'll go to the authors for conveying anything that they'd like to just at the outset so i'm daniel and i'm a postdoc in california i'll pass it to awful work in the air modern collective intelligence i founded and coordinate the kerosene research laboratory on building inactive models for navigating the anthropocene thank you let's go to stephen and then dean hello welcome my name's steven syllett i'm based in toronto and uh looking forward to hearing the new developments in active influence this paper covers great and then good morning i'm i'm dean i'm a retired guy i live in calgary and i'm trying to see what uh new insights can be provided by all this math because it's quite impressive to me cool so on to lance and conor whichever one of you would like to go first and sort of give us the give us the

big picture and how we got here and how you got to this work before we jump into the paper do you want to go uh you you should go headlines right um so so hi everyone and thanks for inviting me daniel so my name is lance and i'm doing a phd in math and neuroscience in london with uh cal pristine and greg pavliotis and basically i i came to writing this paper and doing this kind of research because i'm interested in the question of artificial intelligence and i think a lot of people who are doing machine learning are approaching the question from an engineering perspective but i'm more interested in the scientific perspective the scientific question of what intelligence is what intelligence is and what do we really need to to create an intelligent system and so um researching around what what people were doing i i found about i found out about the free energy principle and to me was um the most mathematical piece of work that i had that i had seen in neuroscience and it seemed like it could encompass many things and he also had applications in active inference to model the brain and ungenerate autonomous agents and so now in my phd my goal is to deeply understand the free energy principle to to lay out the mathematical foundations because i think the free energy principle started out as an intuition and gradually people have tried to to make it more formal and grounded in mathematics um and so this is what we're trying to do in this paper so it's not this paper is not the end but i think it's a step forward so i'm really excited to to share it and ex like talk about it with you guys awesome sounds really cool about starting as an intuition and then working to ground in mathematics or or then would it be grounded in intuition and then elaborated with mathematics so connor yeah thanks i'm happy to be here um and good morning good afternoon to some of you depending on the time zone so yeah my name is connor heinz i'm a phd student at the max punk institute for animal behavior the department of collective behavior in constance germany and i'm co-advised by carl fristen at ucl and then ian cousin who's my advisor in constance i'm in the u.s at the moment uh but yeah so i i'm also generally interested in bayesian mechanics i think so lance and i started our our phds at the same time and ever since i found out that lance was also interested in kind of the mathematical foundations of the free energy principle we've been in discussions about bayesian mechanics for a long time so i was really grateful that he invited me to contribute to this paper and kind of it relates to my own interests because i actually used to be more involved in neuroscience now i've moved more onto the idea that any generic complex system can be described as engaging in a kind of process of approximate bayesian inference and so i'm exploring that in my own research in the context of collective collective behavior so in particular things like collective motion like schooling and fish and flocking and birds so this paper i think it gets at

kind of the mathematical foundations of sentience and autonomy in all kinds of systems not just nervous systems so that's why the the framing in terms of stochastic processes i think is really powerful because it allows us to extend the framework to all kinds of complex adaptive systems not just the ones we might associate the predictive coding free energy formulation with like brains thanks and a really nice point that like there's activity in the nervous system but it's not moving or the subunits are not moving around so what are the similarities and the differences or what's the framework that's going to allow us to accommodate systems that are moving around like the schools of fish flocks of bird and then systems where the topology is either fixed or slowly changing we have to do similar kinds of dynamical analyses so also welcome dave so stephen i see your hand raised go for it and then anyone else who wants to ask a question at this stage yeah just i'd like to comment i think that was thanks for the introductions and those insights i think um you know as there's been this progression oh we've gone from bayesian approaches and how we had predictive processing approaches and active influence approach approaches it's it's nice to also think about okay once we start to generalize it can be nice to go back um to the bayesian for idea more broadly and see okay now the active influence supports all of that you could then start to open the net so to speak so i'm interested in a lot of community psychology approaches even interactive theater and immersive theater where you're trying to work with people's sense-making so i like that re-broadening to maybe a more heuristic mindset so i thought that was interesting cool so anyone can raise their hand um if at any point but let me just maybe start with our as we tiptoe to the edge of the paper like what brought you either of the authors to this framing did you find something in the junkyard that was looked over or was there something that was already being worked on and you wanted to introduce a new thread or how did it come to be composed and worked on in this way well so that there has been um so i think kel fristen has been working like for the past decade trying to put mathematics uh on the free energy principle and trying to explain why so as i mentioned before i think the free energy principle started out as an and i think he has been working for the past decade in on a mathematical theory or a physical theory for why the free energy should hold um as an explanation for what the brain is doing what we wanted to do in this paper is to um to actually take take these developments uh that were mostly coming from physics um and that weren't necessarily as general as they could be or as mathematically rigorous as they could be and actually put it on a fair mathematical framework and try to explain it clearly and i mean as clearly as possible so that was the goal here so what do you think that intuition was or maybe there's a lot of ways that i'm sure fristen and others have said

it but what is the intuition that at least you are pursuing with this work right um so i think the the intuition so i think it came it came from two two different ways um from from two different ideas so so the first idea is if you come from machine learning and you look at machine learning from a very broad perspective you look at many different algorithms and you look at what they do you look at many different problems and you look at what they do and it actually turns out um and i think that's something that people started to realize in the early like 2000s that a lot of these algorithms arguably all of these algorithms they can be seen as minimizing free energy and so we without getting deeply into that um i think the moral of the story is that a lot of the things that working in machine learning are special cases of minimizing free energy so of doing some kind of base and and so many of these problems you can think of them as like special cases of base and inference so i think if you come from machine learning then would be quite natural to say or at least you could say oh approximate inference or free energy minimization is the most general thing you can do in and therefore since the brain engages in some kind of cognition it seems reasonable to assume that it may be doing some kind of basin difference an approximate base inference so minimizing free energy is the most general way to look at it so that's one way to get to a free energy principle but maybe it's not very convincing so i think the other the other way you could get to a free energy principle um which is more grounded in physics and i think that was like um the content of the original papers is that if you look at a system that persists over time so a system that does not dissipate then you can show that it has to be optimizing something and there and then you there has to be optimizing some what people call model evidence so the this system must have like uh implicitly or explicitly a model of its environment and it has to be optimizing in order not to dissipate it has to be optimizing that model and then you can show that actually minimizing free energy enables you to do that so i think that's the the other way um anyway so so what we did in this paper is we we took this intuition of what it means to persist so so we there's like actually a clear way of thinking about those things in uh in mathematics and stochastic processes so what we do in this paper is we look at systems that persist over time we also look at systems so that's the first assumption and then the second assumption is that the system has to be separated by a boundary with its environment and i think i mean intuitively it makes sense because we we're looking at systems that are bounded uh with their environment so they interact via a markup blanket but they cannot mix per se with the system it's not like milk and coffee if you wish it's like a person in an environment so it's well defined in space um so these are the two things that we assume in this paper and then we show that

you can actually describe the system as minimizing free energy in some way thanks for that it's interesting about the machine learning branch towards the fep and then the physics branch because of course there's also sort of the third leg of that triangle with the physics and machine learning so we go from that just a interdisciplinary connection or an interface to add in the third which is this active inference or the free energy type framings and all of a sudden some of the ideas that were a little bit hard to to map from machine learning and physics kind of come around the horn and maybe there's a way that we can see active inference playing a role there so again anyone can just raise their hand if they have a question or write a question in the live chat here's the road map so these were the sections of the paper which introduced that boundedness notion of the markov blanket discussed basic mechanics active inference and stochastic control and then had a nice discussion and conclusion with a few supplemental sections so what went into how you laid out the arguments or what sections do you think were um condensed or which ones would you want to expand on more so first just to the authors on that and then avo your question hello lance and [Music] wait commer i have a question uh before we get into detail i'm curious whether you were aware of the kind of criticism that an activist launched at the french principle in suffocation in cognitive science right could you mention what what specific criticism we're talking about okay so i'm referring to the paper laying down a path by di paolo thomson and beer um there is a lot of criticism that is just completely out unrelated to anything that uh acting says but the core thing that seems relevant to me is the role of creation and historicity well self-creation in cognition intelligence and life so yeah the paper just i'm going to put it in the chat is laying down a forking path incompatibilities between an action and the free energy principle so first to the authors for any thoughts on that and then see stephen with a hand raised right um so so i haven't read this particular so i i can i can't give you an answer that's specific to this particular paper um i would say i would say from a general point of view so yeah there's a lot of criticism on and i think it's natural because the free energy principle aims to be seems to be very general it's also an extremely ambitious um in in regards to to the criticism so far it's it's either been on the on the mathematics and so this is something that we've fixed with the with the current paper so i think a lot of the mathematical criticism in the literature is void i mean is rendered void by by this paper and there are there are other um i would say philosophical criticisms uh on the free energy principle so it's not something i'm expert about because i'm i'm a mathematician so i wouldn't be able to to really debate about the philosophy here um i i so i i'm not aware of like uh daniel you said you put the link to the paper in the chat oh i'll put it in this i'll put it in the

jitsi chat in just a second okay because i i meant i noticed you mentioned the words like creation and historicity um i'm not aware of the particular argument but an argument that i've heard against the free energy principle that reminds me of what you just said and this is one of the more philosophical arguments not the mathematical things that lance just mentioned is the idea that the free energy principle the inference interpretation of a system with a markup blanket at steady state it rests on the idea of a steady state so basically there's an invariant measure of the system over time and when people have tried to say relate or claim that something like societies or large-scale socio-cultural or socio-technical systems are engaging in free energy you kind of have to enforce this assumption of a stationary uh a stationary steady-state distribution on the system and so when you say things like historicity i think of like revolutions or something right so that kind of things where the the kind of intrinsic nature of the system seems to shatter or fundamentally change and that would seem to question the or violate the steady-state assumption which is kind of a baseline thing you need in order for the free energy principle maths to apply um i can respond to that argument but i don't actually know if that's the argument laid out in this paper that's just kind of what my brain jumped to when you mention things like historicity creation and destruction is that is that the the core of the argument though or uh so thing like a revolution is not necessarily a sign of historicity it can be a predetermined in the um in the formulation of the dynamical system but uh what i'm speaking about in the medical terms is the presence of the of difficulties that will be reversible if for example uh we have we drive on the uh where am i right of the road because uh some guy said so like 200 years ago and this does not leave the law ever this is an example of uh walking or path dependency or historicity uh so that is the kind of argument that di paulo and thompson and beer are averaging but it is um formally not proportionate because advocacy does not entail the absence of historicity it's bernoulli systems that do not show historicity ergotic system only do not show like irreversible transitions which is a much quicker condition but where there is a convergence to a better argument is that to have a space or description of a dynamical system you have to assume as you assume wait english is broken assume that this space is fixed throughout the evolution of a system and if we agree with an activists and a system biologists like kaufman and montebell and metro monteville core property of the integrity system is to uh well make sense uh to find observables and to create solutions to problems did not exist beforehand then the um the stat space is not fixed it's created along the systems evolution this is not something you can account with a stat space description of chemical system uh that has a state attractive state distribution sorry yeah i i think i would

agree that as it currently stands the free energy principle couldn't deal the math couldn't deal with um varying state spaces where literally the actual interpretations or like the the finite dimensionality of the system states is changing over time usually you assume some fixed integer dimension for all the stochastic processing stuff although i'm maybe lance could comment on that i'm sure that could potentially be extended to infinite dimensional like partial differential equation style formulations but uh based i if i understood you correctly the first part of what you said about path dependence and irreversibility that is all totally acute uh accommodated by the maths laid out in lance's paper so there's a very you know strong part of the flow that is irreversible another way of describing that as you could say there's a strong degree of path dependence in the trajectories of the system so path dependence per se is totally accommodated and easily um you know path dependent systems can can fit into the free energy framework but the state space changing itself over time that's something that i haven't really even thought about um it's an interesting argument but i i would agree that i don't think the free energy principle currently is equipped to deal with like the state space itself being altered over time that's usually assumed to be fixed but i don't know maybe lance has a different opinion on right um yeah i completely agree with conor so in this paper we don't discuss changing state spaces and i think in terms of future work that would be really really the next step um so so i think the the current formulation uh gives a mathematical framework to think about adapting systems but it doesn't it doesn't really talk about like environments that structurally change over time for example thanks um steven and then we can continue on on this i think this also speaks to that the legs of that stool that was being talked about earlier if you're coming from a math and machine learning you've got you know you've got more lower dimensional approaches but they show the principles and then when you get into the inactive approaches they're looking more from a biological perspective so it might also be a little bit around the very high complexity but the my understanding is that the dependency and ergodicity and also the ability to do approximation uh interpretations of the machine does start to make this math overcome some of those limitations um because it it has more i i i that's my understanding is that the the reliance on ergodicity is not as rigid now so let's go to the markov blanket and then return to some of these important distinctions like ergotic systems steady state stationary so i'm gonna put the uh markov blanket up through time figure three um figure one has just sort of the the timeless the classic blanket and then figure three is adding a subscript t so having that uh those variables describe time dependent functions since that's kind of what we're talking about with the dynamical systems so maybe to the authors like how

do you summarize what this graphical figure means what is it being what is being blanketed what does it mean to blanket and then how does that relate to steady state or stationary distributions which are in the title of the paper right so the the mark of blanket is really the way the mathematical way in which you you talk about boundaries and so having a markup blanket simply means that the internal states here they're separated by they're separated from the external states by the blanket states and then the fact that uh the system is at steady state uh means that the the blanket and so this is really the two things that we need when we're talking about um a biological system or or a stone so we need we need something internal to the stone or the biological system something external and then some kind of interface and of course when you're talking about something very simple like a stone it's not very interesting because you don't have any exchanges between what's internal and external via the blanket but when you're talking about biological systems like like cells or or bacillus you do have exchanges of matter of energy and information and that's something that you can accommodate with the markup blanket that's it's almost a tension in the definition of boundedness which is you said well there is a boundary but then also it's exchanging matter energy information so what is the boundaries meaning uh if those are allowed to pass through it conor and then anyone else uh i'm unmuted right no no you're good i'm good okay yeah yeah so i would uh i would say that's a great question because basically the the boundary we're talking about isn't when you take a boundary for like a physical system people often make assumptions about causality so that something on one side of the blanket can't affect something on the other side of the blanket but because of this exchange of matter and energy across the blanket the the actual definition of the markup blanket is a statistical one it just means that this boundary statistically in a sense segregates the internal on the outside and and that statistical separation can actually critically differ from just direct causal interactions so the actual definition of the markov blink in a statistical sense is that the internal states when conditioned on the blanket states are statistically independent of the external states and vice versa the external states when conditioned on blanket states are statistically independent of the internal states and just saying that definition might not make the intuition any more clear but that really is all that it is um so then when we actually look at the dynamics of particular physical or systems or stochastic processes that statement cannot entail all kinds of causal graphs and all kinds of influences between those subsets um it's really the statistical independence which which defines the boundary um yeah so i think that's that's kind of the kernel of the definition and for certain kinds of statistical distributions we can actually really precisely look at the structure of things

like the stationary covariance which is basically the covariance matrix of the process at steady state and you can read off what systems are outside the blanket what what subsets of variables outside the blanket what subsets are inside the blanket which subsets the variables are the blanket um so it's a really precise definition that kind of goes beyond whatever intuitions you might bake into that on your own about causality awesome points so survival sorry does this not entail that blanket if there is any causal interaction between internal and external states then blanket states would mediate this causal interaction because if they did not then you would have information that will go directly from external to internal or vice versa so um settlement is correct then there is some causal sense to blankets is it correct or not yeah yeah definitely and that's a really good point um so really what the blanket does is uh to to separate the inside from the outside statistically and so what that means is that any information about the internal states that any transmission of information on anything about the internal states that influences external states has to do through the blanket states um so there is a notion of causality but but it's like they as you said the internal and external cannot affect each other directly they have to do it through the blanket okay thank you awesome so dean and then stephen yeah this might be a strange question but that's fine so the blank so the blankets kind of we just use that as a way of describing the interface my question is and i cuz i most of the math that you guys use is is over my head but i appreciate that you've been able to come up with it my question is we're thinking about this metaphorically does the blanket come with its own perforations i mean we talk about it acting as a mediator but is there a math that can kind of show it's it's it's managing this information flow back and forth but does it already exist with these perforations present or is the how is the math going to tell us how that that medium works or is it did you guys already explain that because maybe i just didn't appreciate it when i read the paper yeah that's a really good question so we didn't expand on that in the paper um and so it's good that we talk about it now the definition of markup blanket as it is taking the paper allows for boundaries that are completely opaque so without any kind of transmission but also boundaries that would let anything through so it with this formalism is is quite general um and so far we don't make any assumptions about the structure of the blanket and so it could be very permeable not very permeable you could have all sorts of things um and and i think that that's also really interesting because then i mean ultimately we want a theory that describes biological systems and so the blanket states have some degree of permeability in in biological systems that we can actually exploit mathematically so we don't we don't do that here here the blanket is completely arbitrary and so you could describe a stone but it could

also describe maybe a brain or a rat or something like this but then in the future would be really interesting to actually impose some more structure on the blanket to see whether we can say anything more interesting about the system that we're looking at thanks steven i think this is an area that's probably going to be growing a lot is how much of that transfer process of information from the infinite almost external states gets onto a blanket and then gets transferred to the internal states so the question is normally everyone talks about changing the generative model but there is obviously a question what what can you change in terms of the blanket but i think i don't think that's been fully explored yet so but i think that's a big area that i suppose could be seen as part of the generative model or the regime of attention but it sounds like this the mathematical model here is starting to be able to engage that more if it was expanded yeah let's walk through a few steps of the paper because that's really where we're building up this intuition or moving from the intuition and scaffolding it with the math so first you phrase the blanket as intervening states between internal and external states and defined the state space that those different internal external and blanket states can exist in so totally subject to to servals comments about state space um maybe it'd be helpful to explain this big pie so what is the covariance precision what are we talking about with these matrices what's a system with a certain type of matrix structure that might help us get a grasp on what the covariance matrix at steady state is actually representing about systems right so so in the in our in our scenario here the the matrix π actually encodes everything about um the permeability of the branch of the blanket but more generally um how do the internal states interact with blanket states and how do the external states interact with blanket state so matrix Σ really gives you um yeah the structure of the interaction and um and yeah and so so we just simply showed that the mark of blanket condition so this boundary actually is equivalent to some kind of structure of the matrix by but then when we try to differentiate um living systems with non-living systems and then intelligent systems from non-intelligent systems then probably we could say a lot more about this matrix by so like if there was non-zero everywhere in π what would that correspond to or what would it correspond to have all zeros what does a big number or small number mean in this precision matrix because this is truly a key variable in these formalisms so so the so one extreme would be the matrix by to be the identity matrix and then this would mean that uh each state only interacts with itself and so that that's like kind of an extreme case of a markov blanket because you have a boundary but then you don't even have any transmission with the boundary like you have just three separate states external blanket and internal that don't talk to each other and then the the other extreme

would be to um to to have a non-zero non-zero entry in the in the equation that we have there and then it would mean that internal states and external states can talk to each other directly without interacting via the blanket scene so then you don't have a blanket you just have three states that mix with each other yeah you do you lose you lose kind of the the basic point that says oh we're looking at systems that have a boundary um and and then in terms of the the magnitude of the entries so a big magnitude would say um i think i may get get it the wrong way wrong way around but i think a big magnitude is like you have a lot of interaction and if you have a very low magnitude you have very little interaction okay great because this variable is important and this relationship between the one over the precision and the variance is gonna come up um a lot so we go from this idea of using a matrix to represent basically the statistical connectivity of different nodes and then hope that the structure of that matrix is such that it falls out or designing systems such that the structure of the matrix falls out into internal external and mediating blanket states you develop this notion that the that there's an expected uh internal state and external state that are downstream of the blanket states so what is happening with these formalisms or how can it be that the blanket state tells us about what to expect internally and externally so because you have a because you have a boundary because you have these blanket states so so the whole um so let me give you an example uh if you think about a human then for example part of the blanket states would be what happens on your retina so the impression from the world that you get from in on the retina this is to say that if you have some kind of sensory impression on your retina then this gives you some information about what's going on outside um if if on your retina you see kind of like a a certain shape maybe you you maybe the most like the thing that's going on in the environment that corresponds to that shape is a chair and maybe actually that shape that you see on your eyes like looks like a chair so then you would say oh the most likely external state is a chair um so this is all formalizing the fact that the blanket states what what you have on on your retina gives you information about what's going on outside and also influences what's going on in your brain because you need to process that information and so yeah so we have a most likely external state and a most likely internal state any other um uh thoughts on this because then that's where we go from oh yes so connor go for it yeah i'll just add that uh it's critical that we're talking about expectations here so that's why we're writing down like a normal distribution that's centered on this expectation because remember these are stochastic processes so for any given blanket state it doesn't mean that the internal state on the other that directly is downstream of that blanket state will always be that expected one but this is just the

mean of the distribution but the powerful thing about this uh the bayesian mechanics for these uh gaussian processes is that those um expected states are basically just linear functions of the blanket states just given by these sub matrices of the covariance and then their inverses and such like that so it's just important to know that if we're to pursue this example of the retina it's like for any given retinal impression it doesn't mean there's always going to be the same cortical response right in visual cortex or what have you but there's going to be some distribution of cortical responses and what we're talking about right here is the mean of that distribution very good point so steven and then anyone else with a raised hand and then we'll move forward from what is you're saying there with stochastic processes is there's there's some in a in a sort of a random kind of informational sense and obviously it's lossy as it goes through and it's the ex the expectations i suppose at some level have to have have to be able to be confirmed or disconfirmed or find some sort of statistical correlation to then actually use the information that has to be i suppose at some point there's just not enough there to work anything out or the model's not good enough to find it i'll be interested in how much do you have a sense of how much tends to get through or like how long it takes for information to be processable like if that's if there's some sort of heuristic or whether that's a little bit too unrealistic yeah that's a really great question i mean in the in the brain there's evidence that you know you can get pretty complex i'm not a vision neuroscientist but you can get pretty deep in the cortical hierarchy even up to the level of a decision within you know 150 or sometimes less milliseconds um so that's almost kind of so this is another big important thing to discuss about this paper is the information processing we're talking about here is on the level of expectations right so if the brain is really processing like an immediate retinal impression it doesn't have time to average across a ton of no like tons of parallel noise realizations like and then and then use the expected value to actually do its processing this so the the brain in any given moment is actually just getting stochastic realizations and then it's using information processing on those realizations so i would also i don't know how lance feels about this but i would hesitate in reading in like the expected cortical state given a particular blanket state i would i would with restraint from reading that as the informational currency that the brain actually works with so there is information uh transfer but it's not like the brain always has to work with expectations it can work with um stochastic samples at any given moment but that's also like a controversial thing in in neuroscience right now it's like what is the actual information the brain is working with but the fact that the latency on decision making can be so short suggests that the brain isn't really averaging over a bunch of

parallel realizations of something in order to do a computation another piece is that on this slide we're talking about the expectation in terms of the the mean the average value of a distribution conditioned upon blanket states we're talking about average value of internal states conditioned on blanket states we haven't yet introduced this question about internal states expecting or making a prediction about external states so it's almost like expectation is being used um as just the sort of broad statistical concept that you can condition expectation on some other variables or the systems that we kind of care about and are interested in the ones where the internal states are acting as if they are making expectations or predictions about external states which is where it's taken in the immediate following section of the paper that's a really good point yeah what is the sigma function so i know there's some related um work uh like with miguel that we saw a little bit about like what is the sigma what kinds of gaps in the formalism does the sigma cover how did it come to be this way with the mapping function and what does it reflect right um so i would say that the sigma is the most fundamental ingredient to get to the free energy because ultimately we want to be able to save uh what we want to be able to say or or say say that happens or or doesn't happen is the internal states have some kind of representation about the external states uh and so the way to get there is to actually have a sigma function gives that maps from the expected internal state or expected cortical response to the expected external state like what's going on around you so there there needs to be some uh we need we need to find uh like a relationship between internal states and external states to be able to say anything about inference and so what what we do in in this paper is we give conditions under which um you you have this mapping sigma and then you you're able to say oh we have some kind of inference going on i think in in terms of like a historical so in previous papers the the sigma was assumed um and then and then miguel wrote this paper saying oh the sigma uh you need some assumptions to for the sigma to work it doesn't it doesn't always uh sigma doesn't always exist and so what we do in this paper is we we give conditions under which sigma exists and if these conditions fail and sigma doesn't exist and crucially i think these conditions can be verified from data great points here um the synchronization map and a few of these other pieces relate to the idea of an image and a kernel maybe it'd be helpful to learn a little bit about what what are the image and the kernel reflecting i found one representation about mapping the the uh two different spaces to each other using that sort of kernel and image notation but when you're talking about um in lemma 2.1 the existence of the sigma function entailing an image function that maps the internal to the external states like what is an image what is the difference between just the variable in

itself versus an image of it or a kernel of it right so what we're trying to do is to map the expected internal state to the expected external state and so what we mean by image in the first line here is simply that yeah we're we're trying to map variables that could be your expected cortical responses so we don't care about anything that's not your expected political response where so we care about specific type of variables and then we're trying to map these to expected external states so so that's the first line and so then here we we just say that basically the the gist of this result is that if if you have something that's going on in the environment let's say there's a chair in front of you and you get some kind of expected cortical response and then let's suppose there's a sofa in front of you and you get the ex the same expected political response then you will have a problem because it means that your organism cannot differentiate between chairs and sofas because it has the same cortical response for the for the two two different external states and so what we're saying is that we actually don't want that to happen we want our organism to be able to differentiate between different external and so this is um the the condition three and the conditions for uh they're both i mean each of them is equivalent to one and so it puts on the mathematical footing what it means to be able to differentiate between different states in the environment awesome thank you um serval and then we'll ask a question from the live chat for this thing to work you need organism to be able to discriminate between any two states of its environment is that correct um yeah so any two states that we model not necessarily yes i see that mapping like four b1 and b2 being equal to each other then it implies that the that the external predictions are going to be the same yeah so if your expected cortical responses are the same then the expected states must be the um so then again okay this needs to be thought about in terms of representations so the the external world does not need to be model modeled at the molecular level so it's like a model about the external world and so in that model there there's things yeah so so it can be like a coarse grained model of the external world okay so you use the example of a chair in your sofa i'm pretty sure that most of the listening people can do the difference they're pretty sure that most cats cannot well i said a cat i fly it's smaller why would a certain organism develop representation of certain states under the cringey principle um so that that's a really good point and i think to to address that mathematically we would need to to actually investigate a state space that changes over time to actually to actually address the question of how how things learn representations so that's not that's not what uh we haven't done that far that far yet but that's definitely an important thing to to address in the future so what we the the starting point here is you start with a steady state you start with a markov blanket and you have uh so the

representation is already baked in in the markov blanket and and so in the in this result we just say oh if the mark of blanket looks like then you have a good representation if the markup blanket is uh sort of more degenerate then you have a bad so in the in the next slide actually um if you can put up the next slide um so we show that if the if the condition from the previous slide is not met then you then the organism would have a degenerate type of representation where for example it cannot differentiate between cats and between uh chairs and and sofas and and that that i suppose is still fine uh but it's so so yeah it would like conflate complete different external states um so in this paper we assume that this wouldn't happen because then it enables you to like derive results exactly but you could assume that um the organism conflates different things and then you would have some kind of semi-degenerate type of and then everything still carries out but instead of saying oh my organism does exact basic inference and it's like super accurate and so on uh you can only um you can only say oh my organism maybe is doing some kind of approximate inference that's good in some instances and bad in some other instances so i think um you can definitely relax relax that condition and sort of assess and directly how accurate is your representation to reformulate any representation must have a precursor sensory active state in this framework okay very interesting and also we're still in the um inference section we haven't introduced policy planning or planning as inference so it doesn't need to be the perfect state estimator in order to make adequate policy so we've talked about that in many other situations about action and loop and everything so let's look through the paper and then as we build in pieces we can see where these these different segments come into play because we haven't talked about like policy or conditioning inference on policy yet so stephen and then i'll ask the question from the chat one thing that can also help think about expectations and simplify or or get away from the pressure of representations is the the blanket can be seen more like or the perception visual perception is more like an impressionist painter painting dabs on a painting and then stepping out and looking at it and we're just continually like trying to find the textures and the patterns in that might help confirm our expectations rather than it being like we see the shape of the object it could be that as long as we've seen the type of textures um that we're feeling in the space um and that that some of that could be done at the retina level there could be certain things that just gets processed so i i suppose just couching that in the inactivist world of you know really when we say representation that's a bit unloaded in that world so it could be statistically uh predictable in some way yep and we've talked about representation and some of the different views on representation and axial constants uh work on the the armstis and the

representation war so blue's question before we continue on with exploring the formalism blue asked uh do the maths that you have been working with allow for altering the structure or boundary of the blanket itself for example if i go blind that sensory state is no longer available so how do we deal which is a little bit we addressed earlier but there's like different senses that the models can be changing they can be within the box changing at steady state or there are these other types of changes that actually change the structure of the whole boundary or the internal model itself so what kinds of changes violate the model and which kinds of changes are allowing us to continue using the model so uh at the moment um what the what this paper accommodates sudden changes in the blanket so you could at any point change change the blanket and then see what happens so if for example if you go blind you could change the blanket accordingly and and then everything would carry out what this doesn't account for is blankets that continuously change over time so for example if you're looking at an organism throughout its lifetime the organism would after being born it would like grow and age and so on um so we do not accommodate that but we we can accommodate sudden changes in the nine kids awesome thank you and also just anytime people need to drop off is totally cool so let's take another look at that markov blanket persisting through time maybe even return a little bit to what kinds of systems are we talking about there was this very interesting piece in example 3.1 about describing which types of stochastic processes at the gaussian steady state p might fit under this phrasing so several things were mentioned conor how would you summarize this list or is this an exhaustive list is this just the tip of the iceberg what kinds of attributes do these examples share that allow them to fall under this framework and then what violates it and moves us into a different space yeah so this is um so basically the the the core set of stochastic processes that we we deal with in this paper are diffusion processes um so that's just a particular type of markov process which means that basically the current moment is totally uh statistically dependent on the last moment um that's not to say that there can't exist mathematics that uh apply the free energy principle to other sorts of stochastic processes that's probably something that lance is also working on at the moment but um if i was to give like some i mean markov prosthesis are very generic in the sense that tons of biological processes i think and non-biological processes could be cast as as markov processes and i think there's also an important aspect this is kind of touched uh on a little bit in the next section with integrator chains even though you might say oh but like humans have memory there's all kinds of deep temporal dependencies you can have very high dimensional markov processes where with with levels of time scales but at the lowest level i think it's a safe mathematical

assumption to make that things are markov process at the most fine grained level so i think that the these kinds of systems are like pretty unrestricted in the kinds of things they can capture everything you know from financial markets are modeled as diffusion processes the orange dynamic process which is what we use in this paper is a really popular um process used to model things like uh decision making like the drift diffusion process which is used in kind of alternative force choice decision making tasks um and then things like evolutionary dynamics like genetic frequencies drifting over time are modeled as a stochastic like linear diffusion but i should also say that we use orange strength basically linear diffusions in this work and that's how we get these nice linear synchronization maps the mathematics are all linear and easy but uh this also applies to non-linear um diffusion processes so where the drift term basically isn't just a linear map like a b like a big matrix like it would be for an ou process but it could also be some non-linear very you know kind of harsh function um one some other restrictions is that the drift function so basically the flow that determines the motion of the next set of states given the current state that mo that drift function has to be smooth that's another condition um in these kinds of diffusion processes um yeah that is i think i pretty much cover it there's like some technical things like a zigzag process i'm actually not sure what that is that's something to ask uh lance great thanks and what you said about that um the the multiple levels of integration with the integrator chains and about how that addresses some of the comments about historiosity um we can return to that because there's some i think it's really instructive so serval with a question and then anyone else who wants to ask one yeah i have a question which is mathematical in nature for a change i don't get what this means for random nuclear systems to other three states if it is not ergotic because if it is ergodic then due to well erasity you can find a measure that says something about how likely i am to find a system in such or such subspace you should not have the hybridic conditions and you cannot exchange a distribution for time average i don't get where what where the distribution is coming from exactly in the case uh where the system is um kind of deterministic at least yeah i can't think of an example off the moment but there are processes that have an invariant measure where the time and the space averages are not equivalent so if you just look up non-ergotic stationary processes i think there's some examples with like discrete switching processes like um bernoulli variables where the the time and the space averages will actually not be equivalent so that so i think this explicitly taking into account those those kinds of systems um where the time and the space average wouldn't be equivalent so for instance if you think about systems with multiple basins of attraction that might like stochastically jump

between those two basins of attraction the space average there at any given moment will not necessarily be equivalent like a bunch of parallel realizations at any given moment the space average of that might be centered around one of the modes of that process and that will actually be different than the time average which would actually give your distribution over both modes so that's just something that jumps to mind but there are other examples of stationary processes where the space and time average are not equivalent which is the definition of ergodicity okay but uh how do you define this as the steady state distribution then what does it mean and what are the conditions for it to be defined so a steady state distribution basically just means that the statistic the sufficient statistics that define it uh do not vary over time um so they're invariant they're constant basically um so for for a gaussian process the mean does not change over time and the covariance does not change over so that's that's an example um yeah i mean so the definition of the stationary state actually if if you get down to the fundamentals of it i would more say the stationary state is actually just a property of the flow so the flow that the stochastic differential equation itself like the ito style differential equation that's the most fundamental structure of any and the stationary state actually is just something whose gradients drive the flow so rather than um but but that's more an interpretational point but in terms of the actual okay stationary i guess so steady state okay but you have stationary i guess i know what stationary is and you have as a state which is defined if i understand well but distribution p the question is how do you define a distribution of a generical system in the general case of a stationary nickel so the way you define it formally is you define basically the the a partial differential equation that tells you the density of the of the system at every time point and that partial differential equation is called the fock or planck operator and then you basically set the derivative of that partial differential equation equal to zero and then you solve for the density so that's the most like minimal bare bones definition i can give you of the of the steady state is the density which is the solution to the focker plank operator when the focker plank operates derivative with respect to time zero that's the stationary state so another way of saying that is if i start the stochastic process in a bunch of different places and run it over time all those processes will converge to the solution to the focker blank operator which is the the steady state density okay and for what kind of system do we have a stable solution for this rather than say uh i don't have the words original attractor something that switches and uh i don't know goes to uh new places like expensive process um yeah so the precise condition there's basically conditions on the drift that will tell you whether the system explodes or whether it converges so for instance for an ou process

those solutions to the focker plank operator defined when the eigen values of the drift matrix are not too big for instance but what the actual conditions are for all sorts of stochastic processes such that there are unique solutions to the focker plank operator i am i'm not sure i'm not an expert on uh on stochastic processes that's something that lance might be able to give a more rigorous definition to but it really depends on the drift um basically the f function which tells you whether the system has a invariant measure or basically a unique solution and sometimes there's multiple solutions to the focker blank operator which means that you know the system will end up in one thing or another thanks for these helpful explanations just one comment before dean's question is that sometimes the steady state can be at a higher moment so like in the example of a car there's like the position and then there's how fast it's moving and then etc higher and higher derivatives and so the integrator chains and variational inference allow us to find an invariance at a higher level of the distribution that even as the actual position of the car is changing like that a stationary process doesn't mean the car is not moving it just means there's something about the car's movement that we can continue to integrate or derive towards finding a statistical invariance specifically one that's structured kind of like this where it's almost like a diffusion process or a flow across a field so those are some of the pieces that link together and so it's not just that that the position has to be well behaved it might look very non-linear but we're using certain ways of breaking down the problem so that at each level we can frame it as this like linked sequence of tractable calculations connor if you have anything to add on that or dean you can question yeah i think that's a really good point like uh so basically you can correct for a moving frame of reference through differentiation and that's kind of the basis of these integrator chains is something that might be a moving frame of reference if you center it by its frame of reference then it can actually look like something like a stationary diffusion process and you get around that by encoding the system rather in terms of absolute position something like the velocity or the momentum of these higher orders of motion so yeah i just totally agree with everything you said basically and it's like the historical um critique or comment like well that race car has a historiosity it's like it is but its position at the next time point is determined by its position and momentum and etc etc etc now and then that is happening through time so there is historiosity we're not ignoring that but we're setting up the problem with this like vector of higher derivatives such that it actually is being only determined by the next moment so absolutely dean well thanks for stealing all my under there daniel i basically was going to ask so just to make this clear in my head the metaphor that i drew is you've got classical mechanics and

quantum mechanics in the room and then bayesian mechanics is kind of acting like the marriage counselor it doesn't really actually determine what what is the you know the math can't just be precise it just keeps the two other parties in the room and has them work out what's what's a gist and what's a reflection or am i am i kind of way way off in the in the in the reads here yep so the connection to quantum mechanics which was made in this one section of the paper this was actually something that was suggested by by uh greg pavliotis who is uh lance's supervisor imperial i'm actually not convinced that the connection to quantum mechanics is as fundamental like so i i basically i don't think that uh bayesian mechanics is in between classical and uh classical and quantum mechanics i see bayesian mechanics more as the mechanics of statistical distributions or basically the moments of statistical distributions so in that way if i was to put it in between two branches of physics i would probably put it more at the three-way in between classical mechanics statistical thermodynamics and stochastic thermodynamics because it's really uniting the um the kinds of coarse-grained expect expected variables that you deal with with something like statistical thermodynamics with the sorts of motion that you deal with in things like stochastic thermodynamics or classical mechanics so we're talking about the motion of beliefs or about probability distributions and it gets interesting when you have a markov blanket because then you can start imbuing the system with these like quasi-teleological interpretations of one thing being a belief about another thing but the bayesian mechanics per se actually i think it even i mean it's it's it's controversial whether you would call it bayesian mechanics but there is definitely this kind of statistical or belief-based dynamics that you get even in systems without a mark-off blanket even a one-dimensional diffusion process where there's no boundaries or anything you can describe that system as optimizing model evidence or minimizing free energy it's just it's it's unclear how where where the bayesian this is because the bayesian interpretation really comes in when you're conditioning something on observations and then you have uncertainty about another variable that you can only perceive via the observations so that's why the markov blanket i think actually is kind of fundamental to the bayesian slant on the but even without it you have free energy minimization and you have the evolution of beliefs or evolution of belief distributions it's just they're not they're not as they don't have a nice analogy with like representational mechanics or the things we're interested in in cognitive science or biology yeah that's the part where lance was talking about gist and reflection and how a gist morphs into a reflection and that's i don't even know how you would put the math to that but because i don't know i wouldn't know where to begin but that's that's where i thought you guys were

headed am i misled no i think that's right that's kind of um it gets that kind of i think the thing that daniel was saying earlier where there's there's the expected internal state given a particular blanket state but then there's the idea that any instantaneous internal state is actually also encoding an expected external state whether or not that particular internal state is the most likely one given the blanket state so i don't know which of those would map to gist versus reflection but that's basically the idea is that any given realization of the stochastic process at any given moment even if it's off is kind of always parameterizing some just perception about something on the other side of the blanket so the markov blanket the fact that there's a multivariate system with these kinds of conditional independencies that really is fundamental to the interpretation of saying one side is doing some kind of just perception of the other side you just don't have one side or the other without the ability to distinguish them which i think is a good example about how the intuition that there are separable systems is being built upon in this nexus as you described with thermodynamics and statistics and statistical mechanics and there's other approaches to delineate subsystems like synergetics trying to define subsystems geometrically or trying to define subsystems through various ways and maybe those will become integrated as well so steven and then anyone can write a question or raise their hand okay thanks connor for that that explanation is quite useful as well that you know when you mentioned that there can be bayesian inference say on information coming in but the blanket you know gives it gives a way to to hold that information i suppose but it i can see it giving a much more viable route to integrate many sources of information which might be also sort of staggered over time as they come in and they can be integrated i mean you can integrate across blankets whereas if everything's coming in as one mush of you know all data is coming in through the same or or even if it's not directly data but if if you've only got one causal pathway then you can't necessarily i can't see a way that you could disambiguate and do bayesian inference between what we would use blankets for or you could even say multi-sensory integration then that becomes a physical channel so i suppose one question is how would you think about different channels of information as opposed to different blankets in terms of bayesian integration yeah i think the the actual the multi-channel or multi-sensory integration story is pretty nicely accommodated by this and this kind of brings back to an earlier question that daniel had about what is the interpretation of that pi matrix so when you're thinking about the system minimizing free energy another way of thinking about that is it's minimizing prediction errors for each of the sensory modalities so if our blanket states say we have a three-dimensional external state a three-dimensional internal state and a three-dimensional

blanket state because this generalizes to infinite i mean not infinite but very high integer dimension our three-dimensional blanket state you could think of as like eyes ears and somatosensation or something and then the the distinct elements of that pi matrix are basically the weights that the system that the internal states are assigning to each of those sensory channels so if you actually rewrite the flow of the internal states as a function of that pi matrix you can basically see that the internal states has differential sensitivity to each of the three blanket states in fact it's literally integrating them and the weights it uses to do that that sum or that average are provided by the elements of the pi matrix so i think the multi-sensory integration is basically just a higher dimensional blanket state what it doesn't get at with what i've just said is your idea of information from different channels coming in at different time lags or different delays that is what you would have to accommodate with something like an integrator chain where one blanket state is actually a high order derivative of some other blanket state or something like that or of an external state so you can have different blanket states that from the perspective of internal states are all coming in at once but the actual blanket states themselves are like longer scale temporal summaries of other variables one could be the instantaneous temperature the other could be the time derivative of temperature the other one could be the average temperature over an hour or something like that also sometimes i think about it like from a data science example like imagine if we had three variables that are getting passed to us like height weight and favorite color um maybe that reflects this covariance matrix like height and weight up there on the first two have highest correlation with themselves and then they have less correlation with each other and then the off diagonal is not having any covariance we don't get information about the favorite color based upon the other variables say so multi-sensory can just be different state variables that are getting passed through that vector it doesn't have to be that all the blanket states are of the same type the blanket states are just the set of insulating states and they don't all have to be retinal cells now if you were making this type of a model of the retina it would be that way but then if you wanted to make a model of a person who had sound input light input etc you could have the blanket states reflecting multiple different types then you get this challenge of how is the brain integrating different lags and different precisions etc across different modalities so that's not the end of the story but it is um is phrased generally enough in the state space representation such that we can be discussing in the same matrix multiple kinds of inputs yeah exactly yeah actually conor if you had any thoughts on this example like how should we read the you know π equals what and then q equals what right yeah so this is a point where i i kind

of wanted to bring this up earlier because i think i i don't know it would be good i good to talk to lance about this um i think he's coming to the next one so maybe he can follow up but basically this is actually an example of a system where causality does not imply does basically not exactly correlate with the markov blanket so in that pi matrix that's not actually the covariance right that's the inverse of the covariance so that's a really important point the inverse of the stationary covariance is not going to have necessarily zeros in all the same places that the covariance does so you could have two variables that actually have positive or negative correlation with each other as measured by the square root of the covariance that are actually conditionally independent so for instance in this example um the internal which is the first row or the first column and the external which is the third row or the third column they're conditionally independent of each other which means there's a zero in the corresponding entry one three or three so that doesn't mean that the variables are uncorrelated it means that if i condition on blanket states the middle row slash middle column then there's no more information i can get from external states about internal states or from internal states about external states it's basically saying that the internal states get all the information they possibly can from the blanket states and the external states don't add anything else that's actually a different statement than saying that there's no stationary steady-state correlation between internal and external and it goes further than that then if you actually looked at the drift matrix which is the second equation here you can see from the presence of uh elements in the 1 3 and three one elements of that second equation that big thing that's like one one point five two there's actually a direct influence of internal states on external states in the sense that the motion of external states is directly influenced by internal states so if you were to integrate that stochastic differential equation there would be a if you want to call it that a causal connection going from internal to external but nevertheless because of the way that the stationary covariance is designed when you take its inverse internal is still conditionally independent of external so this is a really interesting and kind of counter-intuitive result that is talked about a little bit in miguel's paper it's also talked about a little bit in an upcoming paper that is um in review right now uh that that basically rigorously defines the conditions under which a causal connection actually results in conditional independence between two so that's basically what the example here is showing is that you could have internal directly influence the motion of external or vice versa but in the stationary sense they're still conditionally independent and therefore the markov blanket is still needed to mediate information between them awesome and it was we'll look forward to these papers um but that's a really great way of

laying out how we build upon this idea of just bayesian variables and the graph that describes them to dynamical systems and their covariances through time and then frame that in terms of flow which can then be decomposed into this time reversible and a time irreversible component ties us into this entire flow areas of thermal flow but also information flow and so also just conor while we have you let's take this next step which is let's partition the blanket states into two kinds of sensory on the incoming and action or active states on the outgoing so what is going on with this blanket partitioning how does it relate to active inference how does active inference relate to this bayesian graphical approach more generally yeah so um here we're basically making the move of taking the blanket states and they could be again of general dimension and then partitioning them themselves into two subsets of states uh active states and sensory states so if we if if you basically look at the dynamics of the system and you look at the expected variables you can say that you know the expected internal state given blanket states is doing inference about external states at the same time you can say that the expected active state given sensory states is in some sense minimizing prediction error so basically minimizing the the deviation between the expected internal and the expected external state so that's where you get along the lines of uh of basically an active inference interpretation and i think it's figure six if i'm not mistaken where yeah or no that's predictive processing figure eight is where you can basically show that the active the uh most likely active state so again this is an expected active state is minimizing the sum of precision weighted prediction errors so the the move here is then you relate minimizing when you're minimizing um prediction errors in the context of inference it's like i'm doing optimal bayesian inference when i'm minimizing precision weighted prediction errors in the context of control it's like i'm doing pid control which basically manuel baltierra has shown this in a paper from a few years ago that is the sort of prediction error minimization so now the elements of that pi matrix become the weights that you that you associate to each variable that you're trying to control in like a pid system so pid is this kind of proportional integral derivative control mechanism so we don't do this in the paper but if you were to associate different blanket states with different like derivatives along an integrator chain like the the um integral the derivative and then the second derivative of some process and then those were your blanket states then you could say that the active the most likely active state was basically trying to control the integral first order and then the derivative of a given system trying to control them using a pid control scheme and where the weights that the pid controller is assigned to the integral the first order value and the derivative those weights would be exactly given by the entries of the pi matrix that

correspond to the blanket variables awesome and actually the integrator chain idea really connects some dots between this variational higher and higher derivatives of challenging functions variational inference on nonlinear functions with planning as inference so again it's like going back to that car with the position and the momentum there's one way of working where you just want to fit the function of the car using its again position and higher derivatives but then what does it look like when you need to predict the position of the car so that you can act and intervene and so this just connects between this graphical structure where you can have arbitrary number of layers of nodes so from left to right is going through time but then vertically within a column is in a differentiation an integration relationship so we connect from looking at a vector of position and higher derivatives connect that to the kinds of systems that have uh blanket states that's just really cool to see yeah so with yeah go ahead yeah sorry was there someone else had a question or go for it connor but anyone can raise their hand or any questions in the chat is awesome i was just going to say like it's essentially that it's that um this so if if we were to cast the blanket variables as these different higher this vector of higher order derivatives of a system just by that system conversion to its stationary state it would be implicitly if we started far away from the steady state it would be implicitly essentially controlling each of those variables to some desired set point so that's often the terminology using control theory is that i'm trying to keep the velocity within some range keep the position within some range keep the acceleration within some range so if you interpret the dynamics of the most likely active state as doing pid control basically what it's doing is it's punishing deviations around those desired set points with some control cost where the control cost exactly is given by the pid weight assigned to the acceleration velocity position which is you can read off from the inverse of the stationary covariance i.e the the pi matrix to kind of give a fun example of that um one way of looking at this figure 10 in the helmholtz supplement is kind of like there's an ink blot and so there's some forces that have it dispersing and some that have it staying together and then when you look down from the top that's like the steady state distribution of the ink plot and so that's sort of a diffusionary way to look at it but let's look at this as like a physical hill so the z-score the the height is they're not not the statistical z-scores the height is like um the concentration at steady state of ink so whatever the diffusion processes are it's densest right in the middle and then it gets less so we can think about that as being like the bottom of the bowl and then it gets higher up so when you drop the marble in the bowl it's going to on average go to the bottom of the bowl and then if you were to think about that marble as like a control theoretic agent with its position which is initially not at the

bottom of the bowl and then all of the higher order derivatives so initially if it's stopped on the side of the bowl it's like you're holding the higher derivatives to be zero then you release it and gravity takes effect etc it would almost look like it was like a plane landing on the bottom of the bowl because it would like accelerate and then it would slow down its acceleration and the higher order derivative it would act as if it were doing pid control to converge all of its and then diminish all the derivatives so that it would end with zero velocity and the differential from the mean of the expectation of zero so that's like kind of an inanimate system but that shows how this diffusion field and then the free energy interpretation of kind of like the gibbs free energy dropping into the bowl finding yourself on the bottom of a physical bowl is also related to control theory so there's like so many of these important threads to connect together general comments otherwise i think just we love active inference what is happening in figure eight like what is happening on the right side and maybe why is it not a monotonic uh drop in free energy which is what we saw in the other figures yes so this is um this is basically the demonstration of what you just said that on average the system um and that's a very key point of what of uh just to add that on to what you said on average the marble will look like it's trying to get to the bottom of the bowl and minimize those deviations from the different set points for position acceleration velocity what have you so this is basically showing that so i'll just explain the figure on the left that the white line is basically the most ac the most likely active state which is one of the two blanket states given various settings of the sensory state so the white line is saying for any given setting of the x-axis what is the most likely setting of the y-axis but then we also show overlaid on that expectation which is just a linear line a linear function of the sensory state we also show an actual realization of the process and that's really the the key thing to note here there's a difference between expectations and which are derived from the steady state to statistics versus realizations of the process which are contaminated with all the nitty-gritty messy things of the real world like noise and solenoidal flow and stuff like that so this also goes to answer your second question about the plot on the right is why is the system not doing monotonic decrease on uh free energy and the reason is because there's solenoidal flow so if there is no solenoidal flow and you average this descent across trajectories an individual trajectory even without solenoidal flow might still not do monotonic free energy descent because of noise so the stack stochastic noise you're adding to these diffusion processes can temporarily drive it up the gradients of the free energy but if i average across like a thousand trajectories it would look like a smooth descent on free energy because all the noise realizations would be averaged out and then there would basically just be a

straight line going from wherever you started this ensemble of parallel simulations to the the bowl the minimum free energy but because of solenoidal flow which is this component of the drift that drives the system along the iso contours of the probability density which is not pointing in the direction of the free energy minimum because of that even after you average across trajectories the free energy will not monotonically go down but it'll actually follow some average trajectory that has some curvature to it um so that's an actually interesting interpretation in the active inference sense because it's so showing that on the way to the free energy minimum it's not necessarily going to take the the path laid out by the most likely active state given sensory states it might actually take a more circuitous route and it will end up on the white line by the time it's at the minimum but its journey to that white line on average and in individual realizations will not actually be along the white line um and that's a really interesting there's a connection there to be drawn to the study of basically efficient um diffusion processes that are used in things like markov chain monte carlo so people basically define design diffusions in order to accelerate convergence to a stationary target distribution this is used a lot in like sampling based approximate bayesian inference and there's actually a lot of evidence and there's mathematical proof behind this too that diffusion processes with this kind of curvature solenoidal flow are actually better at reaching the minimum and they do so faster rather than one that's just doing a direct descent on the log probability which is equivalent to the free energy nice thank you one one thing i'll point out about this figure on the left is that the x-axis has twice the uh scale as the y-axis so you can also this white line is like the $y = x$ line like it's right at 0 is that that's the bottom of the bowl it's it's it's just stretched out a little bit in one way but it's kind of like that's just the $y = x$ line that would be if the um that's just the one to one and then it starts up here with the s of t a of t and then it kind of finds its way to the bottom and that connects to figure 10 with the helmholtz decomposition where just like conor was saying we can separate out into this solenoidal which is going like around the isocontour and then this um straight-line component and so the extreme of one case um assuming that such a line of sight exists is like you're at the bottom of the mountain and then you just like straight line up to the top but then the kinds of systems that we're looking at here are where we can decompose their actions into an as if straight line that's always going up the maximum possible elevation gain to the mountain top or the maximum possible descent to the but also there's this wiggle with a solenoidal coupling that's guaranteed to be orthogonal to that first component and that was an interesting point about designing the kinds of diffusion processes that do help us find whether you phrase it as the top

or the bottom but find the free energy optimal place on a landscape that might be very rugged so a lot of misleading elevation changes hey daniel doesn't that just give another feather in the cap for interactionism over instructionalism can you maybe explain that a little more and then we'll see congress yeah i think what conor just pointed out was really critical if you're if we're optimizing we have to be willing to set up condition where we're not enforcing a linear path that we're actually allowing for the um what do you want to call it the degree of variability and that's what interactionism supports so i just i was just listening to that going yeah there's another there's another proof now we've got the mathematical evidence cool it's like work and play a little bit with um the most direct way to get those instructions might be with a little bit of interaction along the way they're just they're um inseparable in the actual realization of the process any thoughts on that connor yeah i think i think that's i mean i'm always excited when people kind of can connect solenoidal flow to specific domains i'm not sure what instructionalism or or interactionism are but i'm assuming that it's basically saying that just a linear classic optimization taking the path of steepest descent is actually kind of maybe counterintuitively not always the best option i think that's just what solenoidal flow captures in any stochastic dynamical system is the fact that there's some curvature uh to the actual dynamics of systems where they're not always trying to go to the mode of the density but they might actually get where you're trying to go by kind of taking an alternative route on the way there and it's just really cool that mathematically you can show that that's actually better than taking the path of steepest descent it's like the ball can't just go down to the bottom in a straight line then put on the brakes the real path takes a little bit of a different route so oh yeah oh i want to slip one more thing in because i i want to give steven a boost before he speaks and that is that i've heard steven a number of times mention the word trust and trust takes time and i think again that interaction and that that trust piece this this math confirms that we don't just have to intuit that there's actually now some some support for what takes time and what will actually optimize maybe the most sought after outcome not always necessarily the best but at least it gives you a higher likelihood of achieving an outcome because you can you can you can now say trust takes time but there's there's a sort of a almost a mathematical imperative to give more time over to something to arrive at the best answer sorry stephen but i just wanted to you know sort of boost what you've been saying a lot and i because i i agree with it thanks dean stephen yeah it's an interesting yeah i think the trust piece and what you could call real life i don't know if we would talk about real life but a couple of questions that sort of come from that is saying is like you know if you're talking about um this this solenoidal flow is is

that property or that um does that become a more optimal way of achieving radiant descent when there's more noise or the more rugged the landscape i suppose that's my first question and the other question is i just wanted your thoughts you mentioned here like active states so this idea of like it's almost like sensory data coming in on active states and we often talk about active states and action states i what's the states of action that starts the organism does but i suppose internally you've just got the active states which are kind of like almost mathematical codependencies so i'd be interested in like i say the amount of noise or ruggedness or approximation that has to happen and how much the solenoidal if it if it really makes a difference or whether the math just is the same if it's smooth and and and just your thoughts on how states are talked about in terms of active or action oh if that's something you think about um yeah so for your first point so when you say the ruggedness of the landscape you mean like it's not just a quadratic single bump but there might be multiple bumps yeah i'm actually not sure i i don't i don't have any intuitions about why what solenoidal flow would do in that case i think like daniel was kind of saying it might actually be better at finding the global optimum the research i'm aware of is just saying that for even unimodal distributions like this sometimes the solenoid flow actually gets you to unique optimum faster even though you would think that direct descent should be fastest in the presence of noise in the dynamics so not a rugged landscape but just the dynamics are perturbed by noise the solar nodal flow will actually get you to the optimum faster but in the case of the actual probability distribution of the steady state is multimodal and rugged itself that's a very interesting question i don't know i just don't know the research what it says and i personally don't have any intuition about whether solano flow would accelerate convergence to the um or not but i i i would be surprised if no one's researched that because it's a very interesting question and then regarding the active states versus action yeah i mean i guess that i haven't thought that much about what like what semantically jumps out to me or what i associate when i hear active state versus action but i guess i think of fluctuations in the active states as actions so as the active state evolves that's what i would call an action or an or a particular realization of an active state as an action and then the active state itself we just refer to as the random variable that's a stochastic process that evolves over time and we refer to that as the active state um so we have the both the two-dimensional in this case two-dimensional blanket states one is a random variable that we call the sensory state another one is a random variable we call an active state so that whole is the active state and maybe individual realizations of it would be an action thanks connor stephen and then anyone else and then also in this last few minutes if anyone in

the live chat has a question they can write it so stephen thanks that that's helpful i think like you mentioned that i was thinking maybe particularly about noise so yeah that makes sense if there's more noise when there's more rugged landscape yeah that's an interesting question that maybe someone's worked on i was wondering what you're basically saying with active states is there are more flux there's a there's a greater flux a bit like sensory state like the information coming in as is more fluctuating and i.e like well you could say lights going at billions of hertz you know or whatever and then it's the the regularities are slowing down as it comes in the active states would be a greater flux compared to going through the blanket and the transferring it to whatever the perturbations that are relevant to our model are on the outside so that's kind of interesting one comment on that is um the actual error dynamics or just baseline dynamics of each of these kinds of states it's going to depend on the system but when we're thinking about biological systems yes the action states can be conditioned on what the system is like if there's one degree of freedom on the elbow joint then it's a dimensionally reduced in comparison to the incoming sensory data for example so active states end up kind of saving the day in a lot of ways which is why it is such a clutch advance in the field to partition the otherwise just co-mingling blanket states into this kind of incoming and outgoing because then we can assign those incoming states and the statistically causal chains coming inbound we can associate that with inference and with everything related to predictive processing and then on the outbound from the internal states going back out to the active states we can associate that with cybernetics control theory and also planning as inference so the partitioning is is a is an immensely important maneuver steven and then anyone else yeah and also the active states become a proprioceptive form of i mean really appropriate perceptive form of sensory input in a funny sort of way like in the sense that how much they change so for instance you push your i'm kind of curious where there's this blending you know you put your hand on a on a hot on a wall and you push you're sensing the wall but you're also sensing your active states aren't changing as much that makes sense because the wall is stopping them there's an interest in but that that all could be happening at this kind of fluctuating speed however like you just mentioned the system structure will the fact that you've got an elbow and the fact that it allows certain things to happen is going to make all that come into context of the system you know so that's quite cool yep um sorry if there's any noise there but yes when the um person is pushing on the wall they might be getting a different than expected or less than expected amount of change in their sensory input wow it's really loud i'm just going to view it for a second incredible times so um both conor and lance have uh departed for today and i believe that we'll

be talking with at least lance next week in the 26.2 so in this last 10 minutes or so let's look back over the paper and any of the other figures and let's think about what we want to ask or what we'd like to explore in that dot too who has any thoughts of this i got i have a thought i didn't raise my hand sorry um at some point the the 60 year old me would like some of this be to be brought down to six-year-old me in terms of the sophistication of the math and sort of maybe walk through some more of those um the ability to be able to explain to somebody that there isn't that there is a an underpinning of math that explains what this medium is about and why things pass through it and and and necessarily allow these two differentiated states to work i know that i know that they have to be able to explain it in in on the level because it's it's complexity so they have to be able to use a certain sophistication of math but if there's some way for them to be able to bring that down to the my pedestrian level my six-year-old level that would be fantastic because then i think it would be a lot easier for people to climb on board with the idea some of this some of this bayesian mechanics stuff um doesn't have to remain remain elusive or out there but it's actually something useful that we can incorporate into a lot of the practices that we would like people to adopt dean agreed good thing to think about and also dave wrote in the chat we seem to be convinced that some kinds of partitioning simplify analysis great point so um internal external blanket is not the only partitioning of how you can partition nodes in a bayesian network and sense an action isn't the only partitioning that is possible for blanket states so different ways of partitioning the problem are going to variously set it up in a way that's going to be confusing or impossible to solve or challenging to solve or potentially easy to solve and sort of the exemplar of the easy problem to solve computationally not the only example but just one example is like again the ball rolling to the bottom of a hill smooth ball perfect gravity convex optimization it's very easy if you know that the shape of the bowl is a quadratic function then all you have to do is just yep go downhill wherever you are really simple heuristic algorithm for getting to the bottom of that quadratic bowl and then we look out on the ruggedness and on the changingness of the fitness landscapes or of the free energy landscapes for real systems and the entire question is again how can we partition nodes or take the derivative of sets of nodes or of different random variables so that we can make that rugged landscape and transform it or at least act on a smooth landscape so that's sort of the broad theme is can we just cut things up and reframe them so that our challenge is more like the marble going down the bowl rather than bushwhacking you know and we're just totally lost and we don't even have a sense of whether we're going up or down and all of our motion is for naught so stephen and then anyone else

yeah i suppose in a way our cognition is trying to help us make things look more and more like things are rolling down a and it's a smoother more clear picture of the world you know as we get older so it'd be interesting maybe to ask a couple more questions about how noise um i know how noise i mean often they use the infomax or softmax the infomax function i think in a lot of active inference modeling you know to add some noise or to add some distributions so i'd be interested in how where does the covariance break down where does it start to become more coherent smoother and just their thoughts on that seems to be quite that seems to be relevant beyond this particular paper thanks dean yeah one last thing to me what i see is is the math now on paper talking about possibility not talking about certainty i still think that what this does is it opens up the possibility this this map opens up the possibility as opposed to arriving at a certain outcome i think all it does is open up that space that's what that markov blanket to me essentially does is it keeps the door open i know it's described as a as a blanket but i honestly think it what it does is it it creates an openness for more math to be able to flow in and so just i know they're not philosophers but i'd like to hear some of their philosophy around what that what that does because i think if people want to say oh well your math doesn't answer all the all the all the questions fine but it does open up a lot of things that previously we couldn't even say we're open to your point so i'd like to find out what their philosophical positions are now that they've been able sort of bring this math to a place where it does prove out cool dean thank you and yes what you said there about like uh from certainty into uncertainty and of course always leaving that room for the interaction it puts everything on the the uncertainty and the precision are there two sides of the same coin the precision is the inverse covariance so if two things are perfectly covariate there's like a one or a high number in that matrix then your precision is high um uh so it's in the math but yes if they're variable with respect to each other then knowing one doesn't tell you about the other if they're not variable with respect to each other then knowing one gives you high precision on the other and it puts all kinds of state inferences on this continuum of precision so pretty awesome conversation really helpful that we were able to speak with the authors and yeah next week in 26.2 we'll look forward to maybe seeing some more of you on the stream having any people submit questions if they want to ask something or just have us digest it or modify a question at all but as you said very important contributions and scales beyond the paper so thanks again everybody and we'll see you next week thanks daniel bye thanks bye